

Post-Incident Procedures

1 System Recovery Procedures

A. VirtualBox Environment Restoration

Goal: Restore Parrot OS VM to a known-good, pre-incident state.

Steps:

1. **Shut down** the compromised VM if still running:
`VBoxManage controlvm "ParrotOSVM" poweroff`

Revert to a snapshot taken before incident:

```
VBoxManage snapshot "ParrotOSVM" list
VBoxManage snapshot "ParrotOSVM" restore <snapshot_name>
```

- 2.
 3. **Start the VM:**
`VBoxManage startvm "ParrotOSVM" --type headless`
 4. Verify system boots cleanly with no errors.
-

B. Parrot OS System Recovery

Goal: Ensure system integrity and remove persistence mechanisms.

Steps:

1. Run full malware scan:
`sudo clamscan -r /`
2. Verify system binaries with package manager:
`sudo dpkg -V`

3. Check for unauthorized users:
`cut -d: -f1 /etc/passwd`
4. Review crontab and systemd services for persistence:
`sudo crontab -l`
`systemctl list-unit-files | grep enabled`

Update packages and reboot:

```
sudo apt update && sudo apt upgrade -y
sudo reboot
```

5.

C. Network Configuration Recovery

Goal: Ensure network is restored securely after incident containment.

Steps:

Reset firewall rules:

```
sudo ufw reset
sudo ufw default deny incoming
sudo ufw default allow outgoing
sudo ufw allow 22/tcp
sudo ufw enable
```

- 1.
2. Verify interfaces:
`ip addr show`

Test connectivity to gateway and internet:

```
ping -c 4 192.168.56.1
ping -c 4 8.8.8.8
```

3.

2 Root Cause Analysis (RCA)

Incident: Unauthorized SSH Login Attempt

Date/Time: 2025-09-15 10:15 EST

Time	Action/Observation
10:15	Wazuh generated alert for multiple failed SSH logins
10:18	Analyst reviewed <code>/var/log/auth.log</code>
10:20	Identified attacker IP 203.0.113.45 with 45 failed attempts
10:25	Blocked IP using <code>ufw deny from 203.0.113.45</code>
10:30	Disabled root login and restarted SSH service
11:00	Incident closed

Contributing Factors:

- SSH root login was enabled (increased attack surface).
- No rate-limiting or fail2ban in place.
- Password-only authentication allowed.

Technical Findings:

- No successful compromise occurred.
- No malware or persistence detected after scans.
- Attacker was performing automated brute-force.

3 Recovery Validation Checklist

Step	Test Procedure	Expected Result	Pass/Fail
1	Boot VM after snapshot restore	System boots without errors	☐
2	Run <code>clamscan</code> full system scan	No malware found	☐
3	Verify SSH configuration (<code>PermitRootLogin no</code>)	Root login disabled	☐
4	Attempt SSH login with valid user	Successful login with key-based auth	☐
5	Review <code>/var/log/auth.log</code> after 24h	No failed brute-force attempts detected	☐

4 IR Process Improvement Recommendations

Based on Lessons Learned:

- **Enable fail2ban** to automatically block repeated failed login attempts:
`sudo apt install fail2ban && sudo systemctl enable fail2ban`

Enforce key-based authentication by editing `/etc/ssh/sshd_config`:

```
PasswordAuthentication no
PubkeyAuthentication yes
```

-
- **Implement scheduled snapshots** of the VM for faster recovery in the future.
- **Create automated firewall baseline script** to restore rules quickly after incident.
- **Add continuous monitoring alerts** in Wazuh for:
 - Changes to `/etc/passwd`
 - New systemd services created
 - Modifications to `sshd_config`

